WW-N-701B
July 20, 1977
SUPERSEDING
Fed. Spec. WW-N-701A
September 22, 1967

FEDERAL SPECIFICATION

NOZZLES; HOSE, GARDEN, (FOR 5/8-INCH AND 3/4-INCH HOSE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers garden hose nozzles of the adjustable periphery jet type.

1.2 Classification.

1.2.1 Types. Garden hose nozzles shall be of the following types, as specified (see 6.2).

Type I - Heavy weight.
Type II - Standard weight.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-Z-363 - Zinc-Base Alloy; Die Castings.
WW-C-624 - Coupling Assembly, Hose (Garden, Water, and Water Suction).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

FSC 4730

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Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Specification:

MIL-H-775 - Hose, Hose Assemblies, Rubber, Plastic, Fabric, or Metal (Including Tubing and Fittings, Nozzles, and Strainers).

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The nozzle design shall be generally in accordance with either the body movement or the spindle movement as indicated in figure 1. Figure 1 indicates the general construction and assembly only. The nozzle shall consist essentially of a water inlet stem, rotatable body, and, for spindle movement, a separate stem spindle.

3.2 Materials.

3.2.1 Type I. All components, without exception, shall be made of brass or bronze.

3.2.2 Type II. All components shall be made of brass, bronze, or zinc base alloys. Die castings of zinc base alloys may be used, provided they conform to QQ-Z-363 as to material workmanship, manufacture, and chemical composition; either composition specified therein may be furnished.

3.3 Components. All components shall be sufficiently heavy to prevent deformation under ordinary service conditions.

3.3.1 Stem. The water inlet stem shall be hollow and shall provide a waterway from the hose to an outlet, or outlets, in the stem, or in case of the spindle movement, in the separate stem spindle. The outlets shall discharge into the nozzle body and then through the discharge orifice. The hose coupling end shall be internally threaded with 11-1/2 threads per inch for mating with couplings conforming to WW-C-624 used on 5/8-inch and 3/4-inch hose. A durable 0.100 to 0.125-inch thick rubber washer of a size and composition to prevent leakage between the stem and hose coupling shall be provided with the stem. The washer seat shall be flat and have a smooth surface. The washer shall not interfere with the water stream flow.

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3.3.2 Body. The body shall enclose the water outlet end of the stem, and the separate stem spindle (when applicable). The body shall be provided with a water discharge orifice. The orifice shall be set back not less than 0.156 inch from the rim forming the orifice of the body.

3.3.2.1 Type I, packing. Durable, readily replaceable packing, secured by a threaded packing nut suitable for preventing leakage between the body and the stem shall be provided. When specified (see 6.2), the nozzle may be furnished with an O-ring instead of packing.

3.3.2.2 Type II, packing. Packing or an O-ring shall be provided that will prevent leakage between the body and the stem. The packing or O-ring shall be secured by a threaded nut or retainer seal. The threaded nut shall prevent accidental separation of the body and stem. The retainer seal shall form the body and stem into an integral unit.

3.4 Discharge adjustment. Provision shall be made for the rate of discharge to be gradually adjusted by rotating the body by hand from a complete shutoff position through the high velocity-wide angle spray-low discharge position, the straight stream position, and the low velocity-high discharge position. Rotation of the body shall either advance or retract the body in relation to the stem, or in case of the spindle movement design, advance, and retract the separate stem spindle in relation to both the body and the stem. The body movement shall be accomplished through threaded connections between the body and the stem. The separate stem spindle movement shall be accomplished through threaded connections between the body and the separate stem spindle. The separate stem spindle shall be fitted to the stem in a manner permitting longitudinal motion but not rotation. The fabrication and finish of the water passages of the nozzle shall be free of burrs, nicks, grooves, or other defects, including lack of concentricity, so that the discharge of water will be smooth, continuous, and well defined.

* 3.4.1 Rotation.

* 3.4.1.1 Type I, rotation. Threads between body and spindle shall be of the proper form and pitch to require not less than 2.75 nor more than six turns of the body between the complete shutoff and low velocity-high discharge positions.

* 3.4.1.2 Type II rotation. Threads between body and spindle shall be of the proper form and pitch to require not less than two nor more than six turns of the body between the complete shutoff and low velocity-high discharge positions.

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* 3.4.2 Longitudinal movement, type I. The longitudinal movement between the complete shutoff position and the low velocity-high discharge position of the nozzle shall be at least 0.438 inch.

3.4.3 Complete shutoff. No leakage shall result when the nozzle is set at the complete shutoff position and subjected to the test described in 4.5.1.

3.4.4 High discharge. With the nozzle adjusted to the low velocity high discharge position, it shall discharge water at the rate of at least 5 gallons per minute (gpm) when subjected to the test described in 4.5.2.

3.5 Dimensions. The nozzle shall conform to the dimensions specified in figure 1.

3.5.1 Length, type I. When the nozzle is adjusted to the complete shutoff position, its overall length shall be at least 3.875 inches.

3.5.2 Length, type II. When the nozzle is adjusted to the complete shutoff position, its overall length shall be at least 3.375 inches.

3.6 Weight.

3.6.1 Weight, type I. The nozzle shall weigh at least 6.5 ounces.

3.6.2 Weight, type II. The nozzle shall weigh at least 4.5 ounces.

3.7 Finish. The nozzle shall be shaped to provide a firm hand grip while it is adjusted and operated.

3.7.1 Finish, type I. Except for suitable knurled bands, the exterior surface of the nozzle shall have a smooth finish.

3.7.2 Finish, type II. Zinc base alloy die castings shall have a corrosion-resisting coating of cadmium, chromium, or zinc chromate, at the option of the manufacturer. The exterior surface of the nozzle shall have a smooth finish. Suitable knurled bands may be provided at the option of the manufacturer.

3.8 Threads.

3.8.1 Threads, type I. All threads shall be machinecut and conform to NBS H28.

3.8.2 Threads, type II. All threads shall conform to standards prevalent among manufacturers who normally produce nozzles of the type specified herein.

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3.9 Marking. The nozzles shall be legibly and permanently marked with the manufacurer's name or with a trade mark of such known character that the source of manufacture is readily recognizable.

3.10 Workmanship. The quality of workmanship shall meet the standards prevalent among manufacturers who normally produce nozzles of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.2.1).
(b) Preparation for delivery inspection (see 4.7).

4.2.1 Quality conformance inspection. Quality conformance inspection shall be performed on the sample nozzles selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. All units of the same type offered to the Government at one time shall be considered a lot for purposes of inspection. The sample unit shall be one complete nozzle.

4.4 Sampling. A random sample of nozzles shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the nozzles shall be based on inspection level II and an Acceptable Quality Level (AQL) of 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

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4.4.2 Sampling for tests. Tests of the nozzles shall be based on inspection level S1 and an AQL of 4.0 percent defective.

4.5 Examination. Each sample selected in accordance with 4.4.1 shall be examined for compliance with the requirements in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This inspection shall encompass all visual examination and dimensional measurements. The nozzles shall be examined in accordance with the classification of defects in table I, and for all defects classified in accordance with MIL-STD-105:

Table I. Classification of defects.

Categories	Defects
Major:	
101	Incomplete or incorrect assembly.
102	Material not as specified.
103	Missing components.
104	Rotation and Logitudinal movement not as specified.
105	Dimensions not as specified.
106	Threads not as specified.
Minor:	
201	Rough Exterior or interior surfaces (e.g. nicks, burrs, grooves, and sharp edges).
202	Weight not as specified.
203	Finish not as specified.
204	Marking not as specified.
205	Lack of concentricity.

4.6 Tests. Each sample selected in accordance with 4.4.2 shall be tested to determine compliance with this specification. Tests shall be conducted as specified in 4.6.1 and 4.6.2.

4.6.1 Shutoff test. The nozzle shall be connected to a hose, fitted with an external threaded coupling conforming to the requirements of WW-C-624. With water discharged under a pressure of 35 pounds per square inch (psi), the nozzle shall be adjusted by hand, without the help of instruments, through the various discharge positions to the complete shutoff position. Any water on the outside of the nozzle shall be removed. After 5 minutes, the nozzle shall be inspected for leakage to determine compliance with this specification. Nonconformance to 3.4.3 shall constitute failure of this test.

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4.6.2. High discharge. Following the shutoff test, the nozzle shall be adjusted from the shutoff position to the low velocity-high discharge position. The water shall be discharged, at a pressure of approximately 35 psi into a container for 1 minute. The amount of discharge shall then be measured to determine compliance (5 gpm) with this specification. When specified (see 6.2), this test shall be performed at any specified pressure up to 100 psi, or if not specified any convenient pressure may be used. Nonconformance to 3.3.4 shall constitute failure of this test. If the test is conducted at a pressure other than 35 psi, the amount of discharge shall be determined by the following formula:

$$D_1 = 0.17D\ P$$

Where: D_1 = Computed water discharge in gpm at test pressure other than 35 psi.
0.17 = Constant.
D = Required discharge in gpm at 35 psi (5 gpm minimum).
P = Test pressure in psi.

4.7 Preparation for delivery inspection. The preservation, packaging, packing, and marking of the nozzles shall be inspected to verify conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging: Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. The nozzles shall be preserved and packaged in accordance with the applicable level A requirements of MIL-H-775.

5.1.2 Level C. The nozzles shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The nozzles shall be packed in accordance with the applicable level A requirements of MIL-H-775.

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5.2.2 Level B. The nozzles shall be packed in accordance with the applicable level B requirements of MIL-H-775. Ten unit packages (100 nozzles) shall be packed in each shipping container.

5.2.3 Level C. The nozzles shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (see 6.2).

5.3.1 Civil agencies. In addition to any apecial marking required by the contract or order, the nozzles shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, the nozzles shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Nozzles covered by this specification are intended for use with 5/8-inch and 3/4-inch garden hose having couplings conforming to the requirements of WW-C-624.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of nozzle required (see 1.2.1).
(c) Whether an O-ring is acceptable for type I nozzle (see 3.3.2.1).
(d) Test pressure, if other than 35 psi (see 4.6.2).
(e) Level of packaging and level of packing required (see 5.1 and 5.2).
(f) Marking required (see 5.3).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423)

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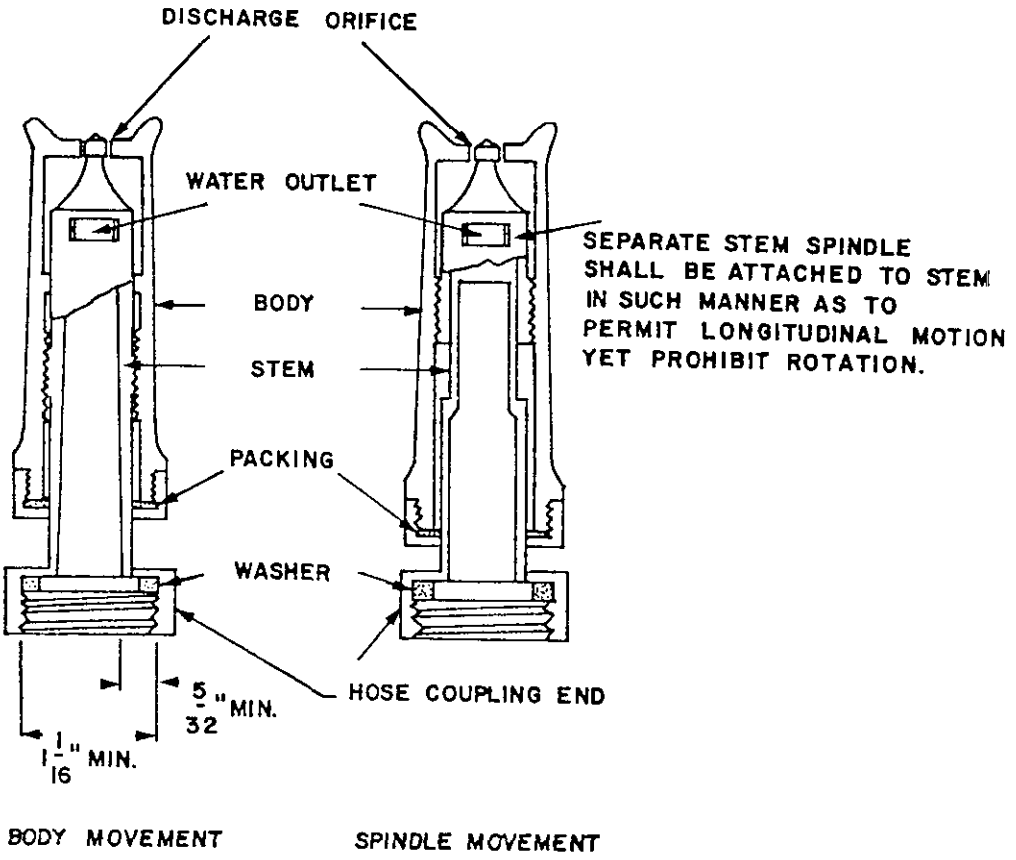


FIGURE 1. Typical nozzle designs. (For general arrangement only)

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incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

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Air Force - 99

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CIVIL AGENCY COORDINATING ACTIVITIES:

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Preparing activity:

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